

Windows 10 Local Backup & Quick Recovery

Restore the hospital HMS and MySQL database rapidly after server crash, disk failure, or ransomware

SCOPE

On-premises Windows 10 + XAMPP MySQL + ZAIZENS HMS Node application

SCRIPTS

scripts/windows-hms-backup.ps1

RTO TARGET

2–4 hours (with tested backups)

RESTORE

scripts/windows-hms-restore.ps1

RPO TARGET

24 hours (nightly) + optional hourly MySQL

Version

1.0 — June 2026

Backup root

D:\HMS-Backups

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1. Backup principles

Objective: After hardware failure, corruption, or accidental deletion, restore clinical operations within hours using offline backups — independent of Railway DR (which is covered in the replication guide).

Layer	What to protect	Priority
Database	tssfsoas_HMS (all tbl_* data)	Critical — nightly minimum
Configuration	.env , config/ , uploaded assets	Critical
Application	HMS_JS deploy folder or build package	High — weekly snapshot
System	Windows image (optional)	Medium — monthly

2. Backup layout & 3-2-1 rule

```
D:\HMS-Backups\
├─ mysql\
│   ├── daily\           ← mysqldump .sql.gz (retain 14 days)
│   └── hourly\         ← optional (retain 48 hours)
├─ config\
│   └── daily\          ← .env, betterpay.json, brand files
├─ app\
│   └── weekly\         ← zip of deploy folder (no node_modules)
├─ logs\
│   └── backup-*.log
└─ MANIFEST.txt         ← last successful backup summary
```

3-2-1 rule:
3 copies · 2 media types · 1 off-site (USB + cloud sync)

- **Copy 1:** Live production on C:
- **Copy 2:** D:\HMS-Backups on separate disk/partition
- **Copy 3:** External USB or encrypted cloud (OneDrive/Google Drive sync of mysql/daily only)

3. MySQL database backup

3.1 Manual dump (verification)

```
"C:\xampp\mysql\bin\mysqldump.exe" -h127.0.0.1 -utssfsoas_Admin -p ^
--single-transaction --routines --triggers --events ^
tssfsoas_HMS > D:\HMS-Backups\mysql>manual-test.sql
```

3.2 Compressed daily backup

The automated script gzips dumps to save space. Verify periodically:

```
"C:\xampp\mysql\bin\mysql.exe" -utssfsoas_Admin -p -e "SELECT COUNT(*) FROM tssfsoas_HMS.tbl_patient"
```

3.3 MySQL data directory (optional)

For advanced recovery, stop MySQL and copy `C:\xampp\mysql\data\tssfsoas_HMS` — only when MySQL is stopped. Logical dumps are safer for routine backups.

Never copy live InnoDB files while MySQL is running — risk of corrupt backup.

4. Application & configuration backup

Path	Include?	Notes
<code>C:\HMS_JS\.env</code>	Yes	Encrypt backup copy — contains DB password
<code>C:\HMS_JS\config\</code>	Yes	BetterPay, branding JSON
<code>C:\HMS_JS\public\dist\</code>	Yes	Built UI bundles
<code>C:\HMS_JS\uploads\</code> (if any)	Yes	Patient documents, letterheads
<code>node_modules\</code>	No	Reinstall with <code>npm install</code>
<code>license-generator\keys\</code> (private)	Vendor only	Not on hospital server ideally

5. Automated backup script

Run from repository:

```
powershell -ExecutionPolicy Bypass -File C:\HMS_JS\scripts\windows-hms-backup.ps1
```

The script:

- Creates dated MySQL dump under `D:\HMS-Backups\mysql\daily\`
- Copies `.env` and `config\` to `config\daily\`
- Prunes dumps older than 14 days
- Writes log to `D:\HMS-Backups\logs\`

5.1 Optional parameters

```
powershell -File scripts\windows-hms-backup.ps1 `
  -BackupRoot "D:\HMS-Backups" `
  -DbName "tssfsoas_HMS" `
  -DbUser "tssfsoas_Admin" `
  -DbPassword "your-password" `
  -XamppBin "C:\xampp\mysql\bin"
```

6. Task Scheduler

Task	Schedule	Script
MySQL + config backup	Daily 02:00	<code>windows-hms-backup.ps1</code>

Task	Schedule	Script
Railway sync (DR)	Every 15–30 min	<code>run-railway-sync.ps1</code>
App folder zip	Weekly Sunday 03:00	Custom zip task (Section 4)

- 1 Task Scheduler → Create Task → name **ZAIZENS HMS Daily Backup**.
- 2 Trigger: daily 02:00, wake computer if needed.
- 3 Action: `powershell.exe -ExecutionPolicy Bypass -File C:\HMS_JS\scripts\windows-hms-backup.ps1`
- 4 Conditions: start only if AC power (for laptops).
- 5 Settings: restart on failure up to 3 times.

7. Full restore procedure (server crash)

Perform restores on a test path first if time permits. In emergency, follow steps sequentially.

- 1 **Prepare hardware** — Windows 10, install XAMPP, Node.js 18+, Git or copy deploy folder.
- 2 **Start MySQL** in XAMPP Control Panel.
- 3 **Create database & user** (if fresh XAMPP):


```
CREATE DATABASE tssfsoas_HMS CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'tssfsoas_Admin'@'localhost' IDENTIFIED BY 'your-password';
GRANT ALL PRIVILEGES ON tssfsoas_HMS.* TO 'tssfsoas_Admin'@'localhost';
FLUSH PRIVILEGES;
```
- 4 **Restore database** from latest backup:


```
powershell -ExecutionPolicy Bypass -File C:\HMS_JS\scripts\windows-hms-restore.ps1 `
-SqlBackup "D:\HMS-Backups\mysql\daily\hms-YYYYMMDD-HHmms.sql.gz"
```
- 5 **Restore application** — copy `HMS_JS` from weekly zip or Git; run `npm install`.
- 6 **Restore config** — copy `.env` from `D:\HMS-Backups\config\daily\`.

7 Start HMS — `npm start` or Windows service (Section 9).

8 Verify — `http://localhost:3000/__health` , login, patient count, print test.

9 Re-enable backups and Railway sync tasks.

8. Partial recovery scenarios

Scenario	Action
Corrupt single table	Restore table from dump: <code>mysql ... < extract_table.sql</code>
Deleted .env only	Restore from <code>config\daily\</code>
Bad deploy / code bug	Revert to last weekly app zip; DB unchanged
Ransomware	Do not pay — rebuild from offline USB backup (not synced folder)
Disk full	Free space; prune old backups; expand partition

9. Windows service & auto-start

Run HMS as a service so it survives logout and reboots (see main Local Deployment guide Section 19). After restore:

- Configure XAMPP MySQL as Windows service (auto start)
- Configure HMS Node service with restart on failure
- Set service account with read access to `.env` only — not Domain Users

10. Testing & runbook

- ☐ Monthly: restore latest dump to `tssfsoas_HMS_test` on same server
- ☐ Quarterly: full restore drill on spare PC or VM
- ☐ Document backup size trend (alert if sudden 50%+ drop)
- ☐ Store printed runbook with Railway DR URL and IT contacts
- ☐ Verify USB off-site copy is unplugged after sync (ransomware protection)
- ☐ After any restore, run `node diagnostic2.js`

Contact / asset	Location
Latest backup path	<code>D:\HMS-Backups\MANIFEST.txt</code>
Railway DR URL	IT runbook / replication guide
Vendor support	ZAIZENS implementation team